Paleomagnetic results of the Cretaceous marine sediments in Tongyouluke, southwest Tarim

SHEN Zhongyue[1], CHEN Hanlin[1], FANG Dajun[1], DING Jinghai[1], ZHANG Shiben[2], HUANG Zhibin[2] & LI Meng[2]

1. Department of Earth Sciences, Zhejiang University, Hangzhou 310027 China;
2. Tarim Oil Field Company, PetroChina Company Limited, Korla 830011, China
Correspondence should be addressed to Shen Zhongyue (email: gs_zshen@dial.zju.edu.cn)

Received January 20, 2003; revised December 15, 2003

Abstract Paleomagnetic and rock magnetic studies on samples of 18 sites from the Cretaceous marine sedimentary rocks in the Tongyouluke section, Akto County, southwest Tarim, China show that the magnetic carriers of the Lower Cretaceous are dominated by hematite with some magnetite, while the magnetic carriers of the Upper Cretaceous are characterized by a combination of magnetite and titanomagnetite as well as hematite and goethite. Stepwise thermal demagnetization is performed and vector analysis is used to isolate magnetic components, which illustrates a single magnetic component or double magnetic components. The high temperature stable components are dual polarities and pass polarity test, reversal test and consistency test. The overall mean direction of the Lower Cretaceous is $D = 27.0^\circ$, $I = 42.0^\circ$, $\alpha_{95}=6.5^\circ$ with pole position at $\phi = 190.3^\circ$, $\lambda=63.1^\circ$, $dp=4.9^\circ$, $dm=8.0^\circ$. The overall mean direction of the Upper Cretaceous is $D = 29.1^\circ$, $I = 39.4^\circ$, $\alpha_{95}=11.2^\circ$ with pole position at $\phi =190.9^\circ$, $\lambda=60.3^\circ$, $dp=8.0^\circ$, $dm=13.4^\circ$. Compared with the inclination of the Early Cretaceous from red beds of north Tarim, the contemporary inclination of southwest Tarim is $10.0^\circ \pm 7.8^\circ$ sharper, but it is still $8.1^\circ \pm 8.9^\circ$, shallower than that of Early Cretaceous basalts in southwest Tarim. Although these paleomagnetic data show slightly larger confidence limit, the paleolatitude of the marine Cretaceous tends to lie between that of terrestrial red beds and volcanic rocks.

Keywords: Tarim paleomagnetism, Cretaceous, magnetic inclination, red beds, marine sedimentary rock.

DOI: 10.1360/03yd0378

Southwest Tarim (hereafter SW Tarim) is one of a few areas that well developed Cretaceous marine sedimentary rocks in China [1]. The Cretaceous marine sediments are stretched in front area along the Tianshan and Kunlun Mountains. Toward the center of Tarim Basin, the Cretaceous sediments are buried by great thickness of Tertiary and Quaternary sediments with little exposure. Compared with the Cretaceous terrestrial strata of north Tarim, the Cretaceous marine strata of SW Tarim continue and develop with abundant fossils that are easily to define the age of the rocks. Researchers from home and abroad have investigated the sequences for decade and a reliable stratigraphic system is established[1].

The Cretaceous is an important period for tectonic evolution in China continent[2,3]. The paleomagnetic data of the Cretaceous achieved before were dominated from terrigenous sediments of Kuqa de-

Copyright by Science in China Press 2005

pression, north Tarim and from basalts in Tuoyun, SW Tarim[4,5], and relatively fewer researches were done on the Tarim Basin[6]. This investigation was focus on the paleomagnetism of the Cretaceous marine sedimentary rocks, which will be a complement of paleomagnetic data for unbalanced space distributions in the westernmost part of China[7]. More importantly, the sedimentary condition and tectonic background of marine sediments in the SW Tarim are quite different from that of terrigenous sediments in north of Kuqa thrust depression, and hence, the paleomagnetic results of the marine sediments of SW Tarim are useful for discussion on red bed inclination shallowing and Cretaceous tectonic evolution in Tarim.

1 Sampling

The Cretaceous marine strata lie in the Tongyouluke section near the Duoyunlike village, Akto County and this section is regarded as a standard profile of Cretaceous marine strata in SW Tarim[1]. The section we sampled concerns about 3000 m long and the geographical position determined with a portable GPS at the bottom of the Cretaceous system is 38°32′10″N, 76°14′27.6″E. The Cretaceous sediments are perfectly outcropped along a dried-up riverside and, from bottom to top, they are composed of Kezilsu Group (K1kz), Kukebai Formation (K2k), Wuyitage Formation (K2w) and Yigeziya Formation (K2y). Systematic sampling was taken to all the formations. Due to the portable driller breakdown in field, we collected oriented samples on rocks with fresh surface in 18 sites, a total of 98 oriented hand pieces. Each site had 3—8 independent oriented specimens, and each specimen was cut into 1—5 samples of 2 cm×2 cm×2 cm cubic form in laboratory, a total of 267 samples for the purpose of magnetic cleaning and rock magnetic measurements. Table 1 lists the sampled strata and the specimens collected.

2 Measurements

All the measurements were carried out in the Paleomagnetism Laboratory, Institute of Geology and Geophysics, Chinese Academy of Sciences. Magnetic remanences were measured with a 2G Enterprise cryogenic magnetometer. Detailed magnetic cleanings were conducted on MMTD-60 thermal demagnetizer that provides program control of heating, cooling and holding, which ensures the consistency of the operations and reliability.

Stepwise isothermal remanent magnetization acquisition up to a maximum field of 2.7T and back-field demagnetization of IRM were performed on a 2G660 pulse magnetizer. IRM of samples was measured on a JR-5 spinner magnetometer. After being pulse magnetized, the Cretaceous gypsum-bearing limestone samples still showed a weak remanence, we have to use 2G cryogenic magnetizer to measure the remanences of various directions in order to ensure a good reliability and repetition for Lowrie test [8].

Temperature-dependence susceptibility experiments using a KLY-3s and its accessory furnace (CS-3) with heating and cooling rate of about 3℃/min were also performed in order to monitor if mineral alteration occurred during heating. Paleomagnetic data were treated by Enkin's paleomagnetic analysis package

Table 1 Sample collected and rock magnetic measurement

Sites	Formation	*n*/*B*/*N*	Rocks	Rock magnetic measurement
T15-T18	Yigeziya Fm. K_2y	31/18/4	Purple gray to light gray massive marl, 50 m thick	IRMs(3) IRM(2)
T14	Wuyitage Fm. K_2w	19/5/1	Variegated mudstone, light gray marl and gray green gypsum mudstone, 124 m thick	IRMs(1) IRM(2)
T12-T13	Kukebai Fm. K_2k	53/11/2	Gray green mudstone, shell limestone, variegated sandy mudstone and gypsum mudstone, 110 m thick	Hy(1) IRMs(1) IRM(3)
T1-T11	Kezilsu Gr. K_1kz	164/64/11	Dark red to brown sandstone and siltstone intercalated with thin-layer mudstone, 1094 m thick	KT(2) IRMs(10) IRM(5)

Sampling location: 38°32′10″N, 76°14′27.6″E; *n*/*B*/*N*: numbers of specimens/samples/sites; Rock magnetic measurement: KT, temperature-dependent susceptibility; Hy, hysteresis loop; IRM, isothermal remanence; IRMs, Lowrie three axial thermal demagnetization; the number in brackets is the number of specimens measured.

and PaleoMag 5.4 package [9].

3 Rock magnetism study

12 samples were measured with IRM acquisition and back-field demagnetization. Fig. 1(a) shows the IRM curve of brown siltstone sample T8-1 of Lower Cretaceous Kezilsu Group. In the field of 0—1000 mT, the remanence is linearly increasing and it is unsaturated at the field of 2700 mT. The remanence rapidly decreases within 50 mT back-field, but its coercivity reaches 750 mT, which suggests that the magnetic carriers of the sample are dominant higher coercivity minerals possibly together with a few lower coercivity minerals, being probably hematite and magnetite respectively. Fig. 1(b) illustrates the IRM curves of the sample T9-2 before and after being heated to 680℃. The heated sample shows a higher coercivity, suggesting that heating causes magnetic carrier mineral change. However, the net magnetic intensity in the oven is less than 10nT and it is not enough to affect the demagnetized direction. Fig. 1(c) and (d) are IRM curves of upper Cretaceous limestone and marl. When the field reaches 150 mT, the remanence is 75% saturated and completely saturated at 300mT(fig. 1(c)). Back-field demagnetization shows its coercivity <100 mT, indicating that the carrier is dominated by lower coercivity minerals. Sample T12-4 shows a rapid drop in very low back-field but its coercivity is as high as 1000 mT(fig. 1(d)), indicating that coexistence of lower and higher coercivity carriers.

The existence of magnetite, goethite and hematite could be confirmed by thermal demagnetization of three-component IRMs. Fig. 2 shows the IRMs behavior with three types: (1) the hard component (0.45—2.7T) is unblocked at temperature about 100℃, suggesting existence of goethite (fig. 2(a)). The unblock temperatures of soft component (0—0.045T) and middle component (0.045—0.45T) are 450℃—480℃ and 550℃ respectively, most possibly related with various Ti-bearing magnetite. (2) Unblock temperature at 320℃—350℃ may be contributed by pyrrhotite, greigite or maghematite; unblock temperature at 580℃ of all three components (hard, middle and soft) illustrates contribution of magnetite (fig. 2(b) and (c)). (3) Unblocking temperature at 675℃ for three components in all samples confirms magnetic carrier of hematite (fig. 2(d)).

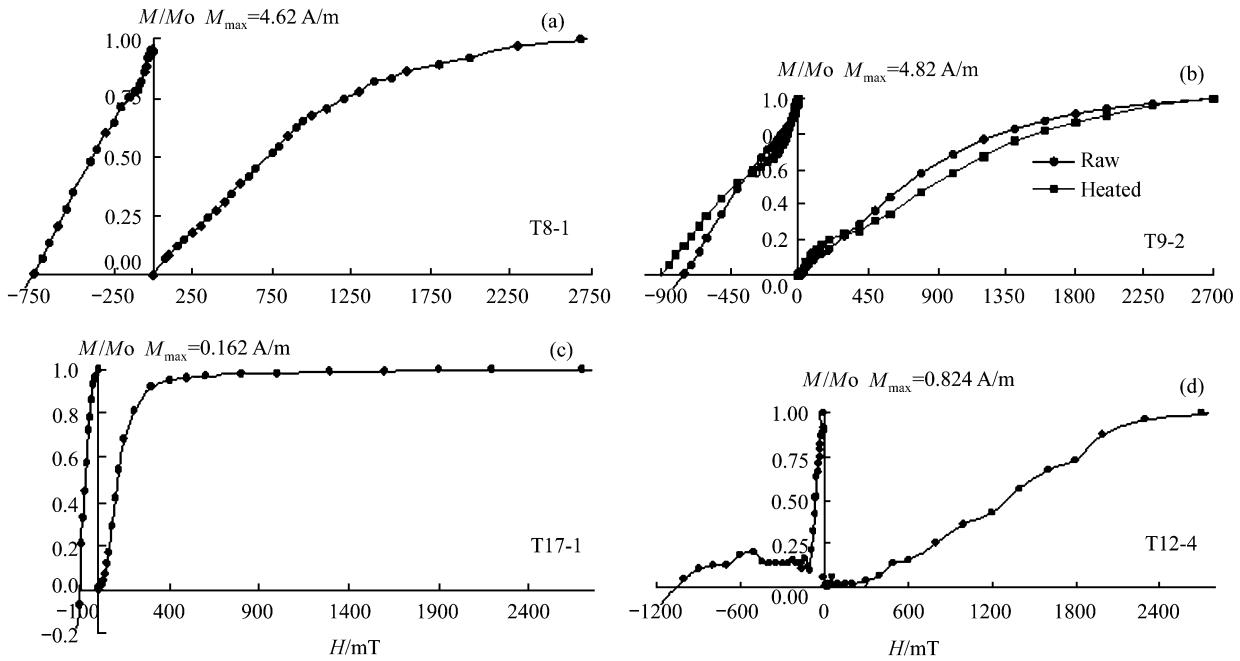


Fig. 1. Typical curves of IRM acquisition and back-field demagnetization. (a) Purple sandstone T8-1; (b) brown fine sandstone T9-2 before and after being heated to 700℃; (c) gray yellow marl T17-1; (d) gray green fine sandstone T12-4.

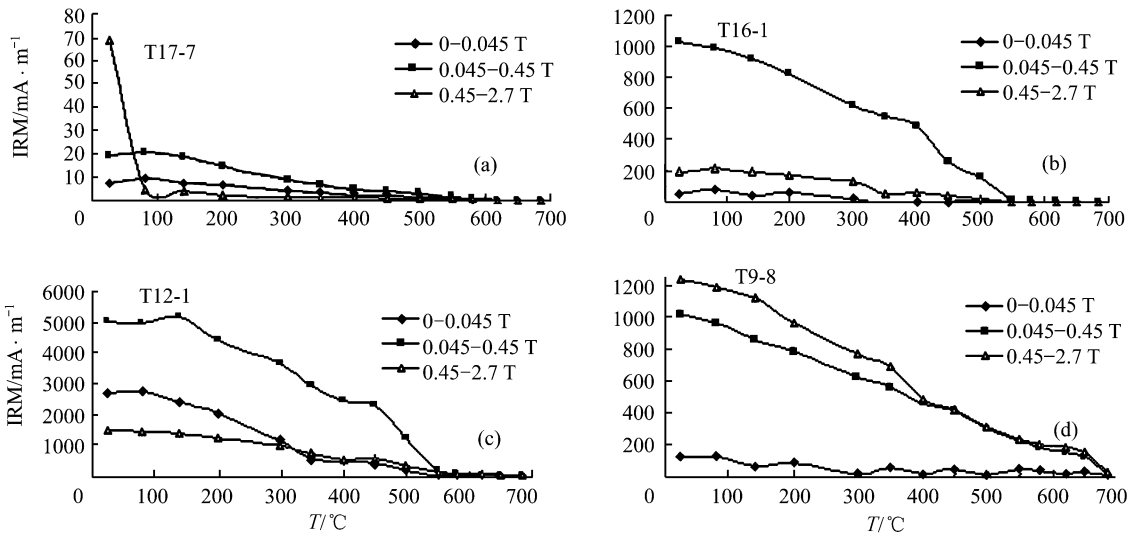


Fig. 2. Thermal demagnetization of three axial composite IRMs. (a) Gray yellow marl T17-7; (b) gray yellow marl T16-1; (c) gray green fine sandstone T12-1; (d) purple fine sandstone T9-8.

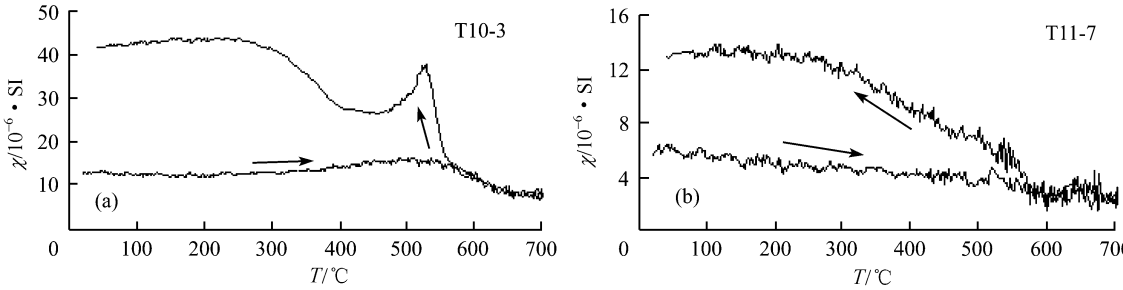


Fig. 3. Typical temperature-dependent susceptibility curves. (a) Gray green sandstone T10-3; (b) variegated sandstone T11-7.

The temperature-dependence magnetic susceptibility curves are shown in fig. 3. For sample T10-3, magnetic susceptibility increases slightly after 350℃, but decreases at 500℃, indicating possibly Hopkinson effect of magnetite. The lowest susceptibility is at about 680℃. However, this run for the Cretaceous sample is not reversible: the cooling curve shows a sharp increase in magnetic susceptibility value when cooled below 580℃(fig. 3(a)), suggesting significant thermally induced formation of new ferromagnetic minerals. For sample T11-7, magnetic susceptibility decreases slightly during heating and a turning point occurs at 580℃, which is identical to the unblocking temperature in IRMs. The magnetic susceptibility decreases to the lowest in 700℃ and shows no obvious evidence of mineral change. The cooling curve reveals a slowly increase in magnetic susceptibility at 580℃, which seems to be probably the change of the magnetic domain (fig. 3(b)).

The rock magnetic study shows that the magnetic carriers in the Lower Cretaceous samples are dominantly hematite, with a small amount of magnetite, while the Upper Cretaceous samples are dominantly by magnetite, with some hematite, goethite and titanomagnetite.

4 Magnetic remanent analysis and remanent test

Stepwise thermal demagnetization was performed on 237 samples with a step of 100℃ in lower temperature and 25℃ in higher temperature. The results

were then plotted with Zijderveld plot[10] and equal-area project. Magnetic components were isolated with vector analysis[11].

Most samples show a simple demagnetization behavior. It is common that an overprint of present-day magnetization or viscous magnetization was cleaned in low-middle temperature and characteristic remanent magnetization (ChRM) was displayed at high temperature. Fig. 4 shows the typical thermal demagnetization results. Sample T8-1 has a natural remanence of 1.979×10^{-4}A/m and it gradually decreases to half of initial intensity at NRM—300℃, complete demagnetization occurred at 650℃—675℃ (fig. 4(a)). The Zijderveld plot reveals a demagnetized

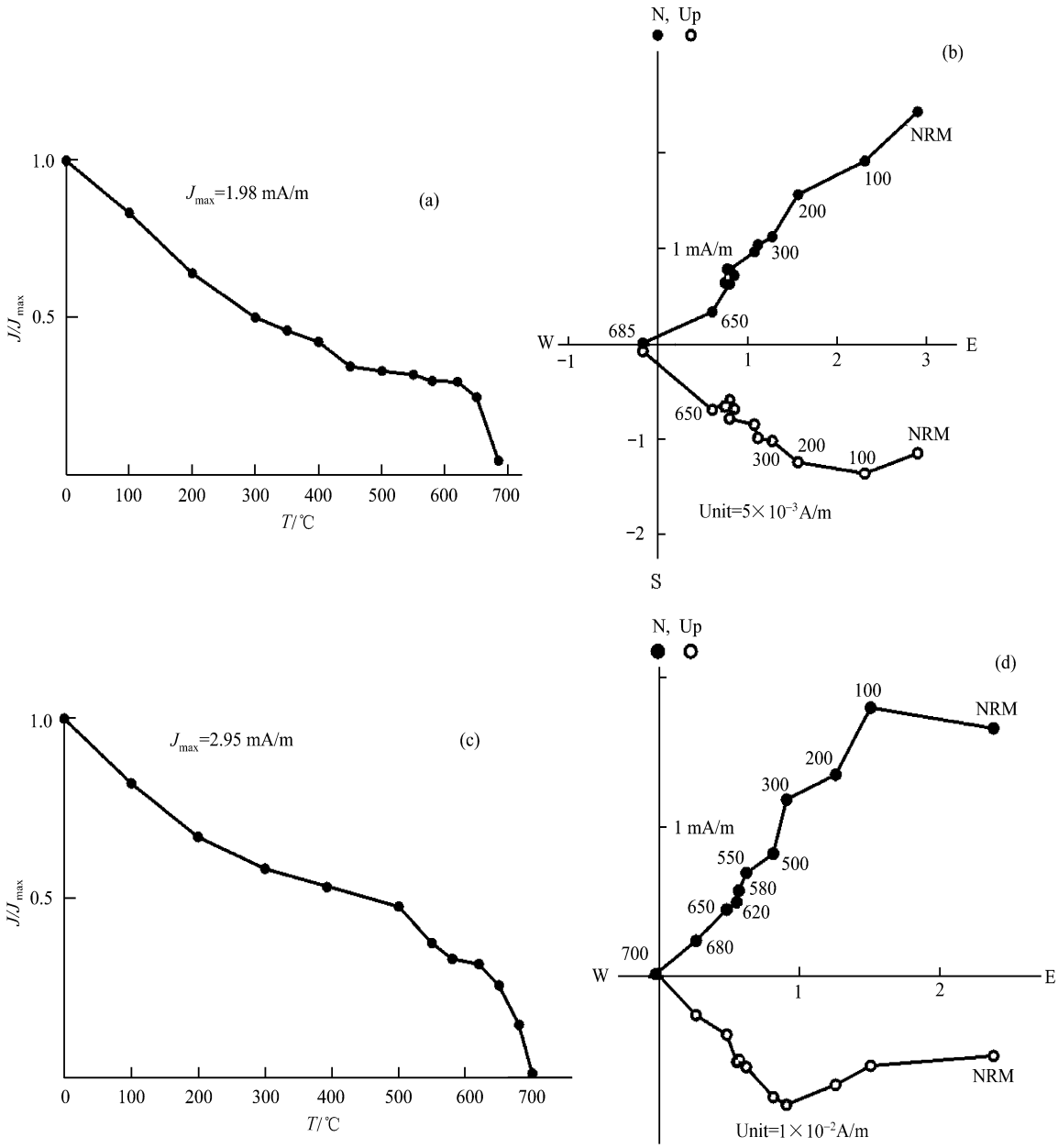


Fig. 4. Representative thermal demagnetization plots. (a) Decay curve of sample T8-1; (b) Zijderveld plot of sample T8-1; (c) decay curve of sample T10-7; (d) Zijderveld plot of sample T10-7.

vector between 100℃—650℃(fig. 4(b)) and the demagnetic vector becomes chaos after 650℃. The final demagnetization is at 675℃, indicating a carrier of hematite and a single magnetic component.

Some of samples show two components. The intensity decaying curve of sample T10-7 shows a slower drop before 500℃ and a turning at 580℃, which is followed by a sharp decay at higher temperature (fig. 4(c)) and final unblocking at 700℃ (fig. 4(c)). Zijderveld plot indicates two vectors at 100℃—300℃ and 500℃—680℃ respectively (fig. 4(d)). The magnetization direction (D_s = 40.8°, I_s=36.8°) at higher temperature is nearly identical to that of the single component samples (fig. 4(b)). Hence, we conclude that the same characteristic components are present in all samples.

All demagnetization results of the Tongyouluke section samples can be grouped into two combinations:

1) Single magnetic component. Magnetic directions basically keep the same between 100℃—680℃ or 580℃;

2) Two magnetic components. Isolated from lower temperature (NRM—300℃) and middle/high temperature (550℃—675℃) respectively. The component of lower temperature is close to the present magnetic direction, reflecting the present viscous magnetization, and the component of middle to higher temperature at 550℃—650℃ for most samples and at 350℃—685℃ for a few samples is a characteristic remanence, carried by magnetite in limestone and gypsum limestone and by hematite in red beds.

Middle to higher temperature component could be isolated from most samples. The site-means of the ChRMs with positive polarity usually are well clustered, while the site-means of ChRMs with negative polarity are relatively dispersed with a larger confidence cycle (fig. 5). The magnetic results of 18 sites could be divided into two groups according to the Early and Late Cretaceous time. In Early Cretaceous, there are 2 negative and 9 positive polarities (fig. 5(a) and (b)), in which negative polarities lie in the lower part of the Early Cretaceous. The site mean direction of the positive polarities is D_s=28.1°, I_s=42.2°, n=9, K_s=42.4, R_s=8.8, while the site mean direction of negative polarities is D_s=202.0°, I_s=–41.1°, n=2, K_s=21.8, R_s=1.95. After being tilt corrected, the high temperature stable components are overlap at 95% confidence level when the reversal direction is inverted, suggesting a single directional population and passing McFadden[12] positive reversal test with a C type classification (γ_o=3.63° < γ_c=12.5°, critical value 0.3949). In Late Cretaceous there are 7 sites with 3 negative sites and 4 positive sites. The magnetic directions of Late Cretaceous are relatively scattered with α_{95}=8.5°—13.2°, still pass I type classification reversal test in 95% confidence level (γ_o=6.16° < γ_c=26.5°, critical value 0.8205). In tilt corrected direction projections (fig. 5(b) and (d)), the positive and negative directions are apparently antipodal. The array of the magnetic polarities over the sampling section is similar to that of Cretaceous marine magnetic polarity time scale. Therefore, this paleomagnetic data could be regarded to pass both reversal test and polarity test. The site-mean directions in geographical coordinates are relatively dispersed (fig. 5(a) and (c); K_{1g} α_{95}= 8.5°; K_{2g} α_{95}=16.7°), but well clustered in stratigraphic coordinates (fig. 5(b) and (d); K_{1s} α_{95}= 6.5°; K_{2s} α_{95}=11.2°). All these reflect that the remanence is a primary one. Unfortunately only a monoclinal occurrence in the studied area, a full fold test is not suitable for this case. The higher temperature components obtained by vector analysis are averaged for multiple samples and a site-mean ChRM direction is then calculated from the specimen ChRM directions. The site-mean and overall mean directions are listed in table 2.

5 Discussion

5.1 Comparison between Cretaceous magnetic pole positions of north and SW Tarim

Fang et al.[13,14] have obtained Early Cretaceous characteristic remanant directions in Kuqa depression, north Tarim. These ChRMs isolated from sites with

Table 2 Site-mean Paleomagnetic data of the Cretaceous from Tongyouluke section

Site	Rock	Age	*n*/*N*	*In situ*		Tilt-corrected			
				D_g/(°)	I_g/(°)	D_s/(°)	I_s/(°)	κ	α_{95}/(°)
T18	Mar	K_2	3/3	89.6	–74.4	219.0	–40.1	28.1	13.1
T17	Mar	K_2	4/5	137.4	–29.4	201.6	–35.1	21.6	12.4
T16	Mar	K_2	7/7	359.4	75.2	38.1	31.9	37.9	8.6
T15	Mar	K_2	3/3	295.9	–77.7	231.5	–26.7	23.2	12.4
T14	Mar	K_2	5/5	307.5	58.4	21.2	40.1	28.8	11.6
T13	Mar	K_2	4/5	263.2	55.2	36.4	49.5	27.8	13.2
T12	Mud	K_2	6/6	287.7	34.4	349.5	41.9	37.3	9.3
T11	St	K_1	5/5	292.9	64.5	27.4	44.9	212.2	4.3
T10	St	K_1	5/6	306.5	62.4	22.3	36.6	51.0	8.8
T9	St	K_1	8/8	307.1	73.5	24.3	37.1	34.7	8.4
T8	St	K_1	5/5	349.2	70.9	33.6	28.7	21.8	11.3
T7	St	K_1	6/6	340.3	76.1	37.0	26.1	26.8	11.0
T6	St	K_1	4/4	148.1	–57.2	188.3	–35.2	115.9	6.5
T5	St	K_1	7/7	272.4	64.7	25.6	56.8	73.6	6.2
T4	St	K_1	4/4	63.9	–80.0	218.0	–45.1	102.8	6.2
T3	St	K_1	6/6	280.8	61.5	27.1	52.9	111.8	5.4
T2	St	K_1	8/8	266.2	69.1	26.6	58.9	96.8	5.0
T1	St	K_1	5/5	171.1	69.9	25.9	37.2	35.2	8.9
	Mean	K_2	7/7	301.9	61.7			10.0	16.7
	Mean	K_2	7/7			29.1	39.4	22.4	11.2
	Mean	K_1	11/11	294.9	73.2			24.8	8.5
	Mean	K_1	11/11			27.0	42.0	41.7	6.5

Age	Paleopole position				Plat/(°)
	$\lambda\rho$/(°)	$\phi\rho$/(°)	dp/(°)	dm/(°)	
K_2	60.3	190.9	8.0	13.4	22.3
K_1	63.1	190.3	4.9	8.0	24.3

Sampling location, 76.24° E, 38.53° N; *n*/*N*, number of samples calculated mean direction/number of samples demagnetized; *D*, declination; *I*, inclination; *g*, geographical coordinates; s, stratigraphic coordinates; α_{95}, 95% confidence angular radius; κ, precision parameter; $\lambda\rho$, latitude of magnetic pole; $\phi\rho$, longitude of magnetic pole; *dp* and *dm*, error of magnetic pole position; Plat, paleolatitude; St, sandstone; Mar, marl; Mud, mudstone.

various locations and rocks are obviously offset from present geomagnetic field in geographical coordinates and dispersed distribution. After tilted correction, they are well clustered. Hence, they are regarded as a magnetization before rock deformation. ChRM components pass the consistency test and generalized reversal test, suggesting a primary magnetization and a representative of the Cretaceous paleopole position of north Tarim (ϕ = 226.0° E, λ = 66.2° N, *dp*=4.9°, *dm*=8.3° [14]).

For comparison, we project the poles from this study and the poles from the Early Cretaceous basalt and red beds in Tuoyun[5](fig. 6).

The Early Cretaceous pole position of SW Tarim is separated from that of north Tarim (fig. 6), while, at 95% confidence level, it is intersected with the pole of red beds intercalated in basalts. Angle between Early Cretaceous poles of north Tarim and red bed of SW Tarim is 15.6° ±6.5°, suggesting relative motion between the blocks of SW Tarim and north Tarim. If we take SW Tarim as a reference point and the north Tarim anticlockwise rotates 15.0° ±7.4°, the two poles would be superposition. Fault-blocks are well developed in the Tarim Basin, especially in the foredeep regions[15]. Under tectonic compression the fault-bocks are locally overthrust or rotated by slide slip motion, which may result in the rotated superposition between the poles of north Tarim and SW Tarim.

5.2 Comparison of the inclination of Cretaceous marine rocks with that of Cretaceous terrestrial rocks and volcanic rocks

For comparison, we calculated the Cretaceous

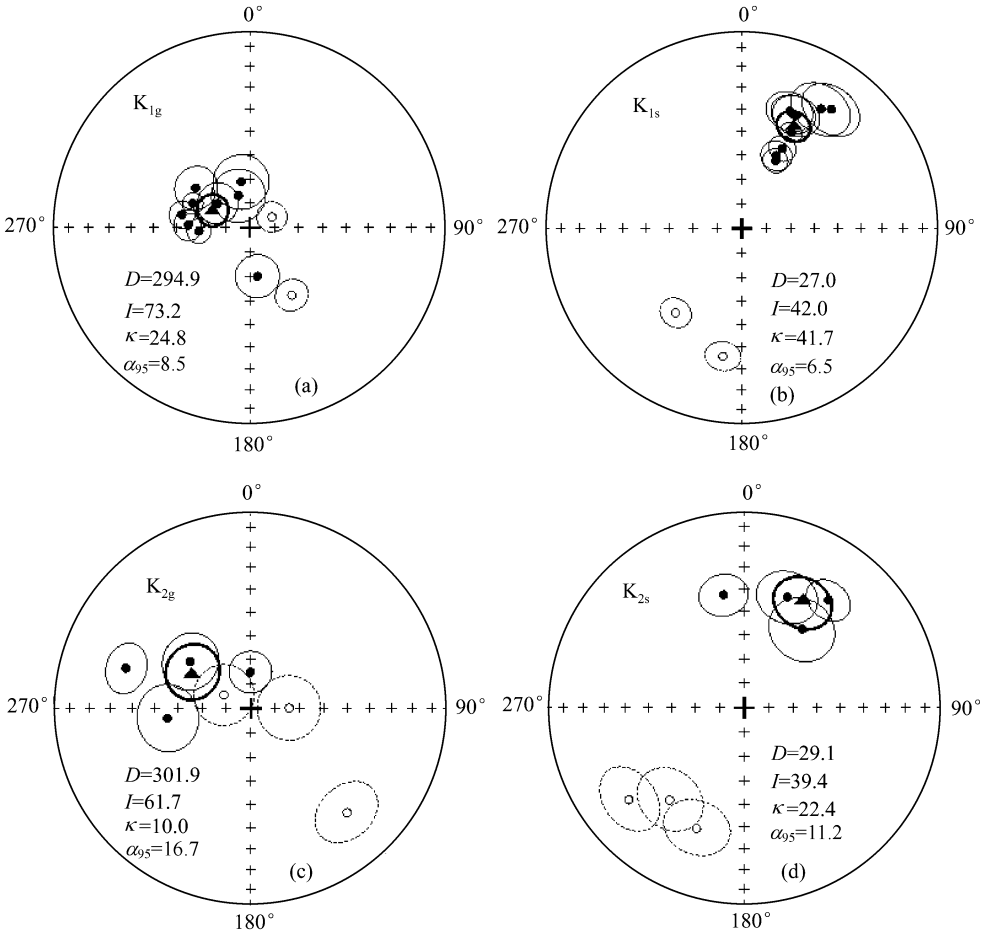


Fig. 5. Equal-area projections of site-mean directions of ChRM together with associated confidence circles. (a) K_1 *in situ*; (b) K_1 tilt corrected; (c) K_2 *in situ*; (d) K_2 tilt corrected. small circle (open circle) and thin ellipse (dashed ellipse), upper hemisphere (lower hemisphere) projection of the site mean direction; triangle, overall mean direction; κ, precise parameter; α_{95}, 95% confidence circle.

paleomagnetic data from Tarim for a common point of Aksu (41.7° N, 80.5° E), north Tarim (table 3). Obviously, the inclination of Early Cretaceous red beds of north Tarim is significantly lower by 10.0°±7.8° than that of Early Cretaceous red beds of SW Tarim. The inclination of Early Cretaceous red beds of this study corresponds with the inclination of the red beds intercalated in Early Cretaceous basalts [5] at 95% confidence level. Despite of the larger confidence limit (α_{95}=10.6°) of the magnetic directions of Tuoyun basalt, its mean inclination is higher by 8.1°±8.9° than those of contemporaneous marine rocks.

Gilder (2003) collected paleomagnetic data of red beds and volcanic rocks from Central Asia (north lati-

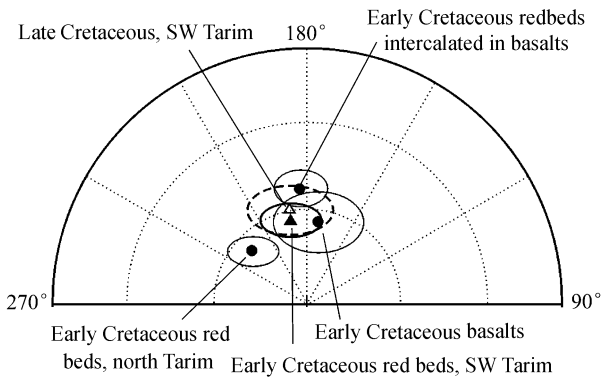


Fig. 6. The pole positions of north and SW Tarim. Data from: Early Cretaceous red beds, [14]; Early Cretaceous basalt, [5]; Early Cretaceous red beds intercalated in basalt, [5]; Early Cretaceous red beds, SW Tarim, this study; thick line for highlighted; Late Cretaceous SW Tarim, this study..

Table 3 Comparison of inclinations of Early Cretaceous north and SW Tarim

Location	Paleopoles			Paleomagnetic parameters (at common point 41.7° N, 80.5° E)		Reference
	$\lambda\rho$/(°)	$\phi\rho$/(°)	α_{95}	Plat/(°)	I/(°)	
K_1 north Tarim red beds	66.2	226.6	7.0	21.0°±7.0°	37.5°±5.9°	[12]
K_1 Tuoyun basalt	64.1	172.1	10.6	36.4°±10.6°	55.6°±7.4°	[5]
K_1 Touyun red beds in basalts	53.2	183.4	6.8	25.6°±6.8°	43.8°±5.4°	[5]
K_1 SW Tarim red beds	63.1	190.3	6.5	28.6°±6.5°	47.5°±5.0°	this study

For abbreviations see table 2.

tude 34.7° — 44.2°, longitude 66.9° — 103.5°) since.

Early Cretaceous and calculated them for a common point of the center of Tarim Basin (39° N, 82° E)[5]. These data are shown in fig. 7 compared against the paleolatitudes calculated from the Besse and Courtillot[14] Eurasian reference apparent polar wander path (APWP). Within uncertainty limits, the mean paleolatitudes from the volcanic rocks are 2° — 4° lower than those from the Eurasian APWP. The paleolatitudes of the terrestrial red beds are significantly lower by 10° —25° and the shallowing has changed with time[5]. The paleolatitude (inclination) defined by the marine Cretaceous sediments of SW Tarim lies between the paleolatitudes (inclinations) of terrestrial red beds and volcanic rocks.

Because of the paleolatitudinal difference between paleolatitudes defined by red beds of Mesozoic-Cenozoic in central Asia and those calculated from Eurasian APWP, many researchers pay attention to this issue[5,13,14,17—19] and put forward several hypotheses to explain the 'anomalous shallowing inclination', such as syn-sedimentation, compaction-induced inclination shallowing, geomagnetic field geometry, tectonic shortening, remagnetization, etc.

In studying the Lower Cretaceous and Lower

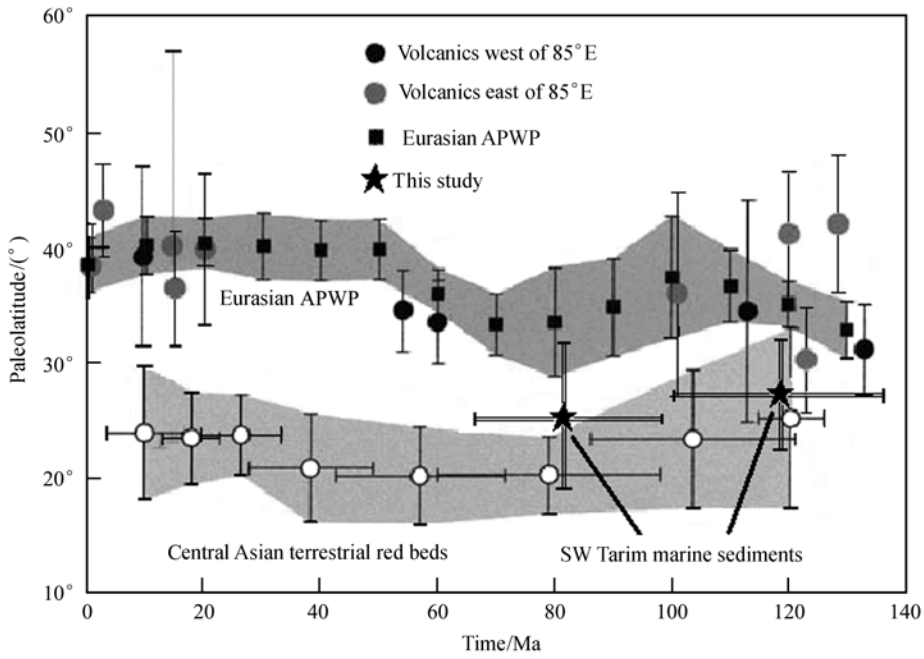


Fig. 7. Paleolatitudes calculated from Eurasian APWP, Tarim basalt and red beds. (modified after fig.7(b) of Gilder et al, 2003). Solid circles, volcanic rocks from central Asia; open circles, terrestrial red beds; squares, the paleolatitudes calculated from Eurasian APWP of Besse and Courtillot[16]; fine horizontal lines, time span; fine vertical lines: the uncertainty limit of paleolatitudes; stars, Cretaceous marine sedimentary rocks of this study; horizontal and vertical double lines, time span and paleolatitude errors of this study. All the data calculated for a common point of central Tarim (39° N, 82° E).

Tertiary paleomagnetism of north Tarim, Fang et al. suggested a shallowing inclination in Cretaceous red beds[13,14,17], and indicated that the compaction is one of the causes that induced the anomalous shallowing inclination[13,17,19]. To explore the possible reason of the Cretaceous inclination discrepancy between north Tarim and SW Tarim, 80 samples were measured with anisotropy of magnetic susceptibility. Magnetic lineation and foliation were calculated. The results show that magnetic foliation (F=1.14) is much stronger than lineation (L=1.03) in samples from the Kuqa section, north Tarim, illustrating a distinct foliation structure. Tan et al. studied the fabric carried by ChRM-carrying hematite particles dissolved during chemical demagnetization and showed an enhanced magnetic foliation[19]. This suggests the Cretaceous rocks north Tarim have been compacted or experienced significant strain due to tectonic stress. While the magnetic foliation (F=1.05) and lineation (L=1.04) of the Cretaceous marine sediments of SW Tarim are weak, and the maximum axis of magnetic susceptibility anisotropy is parallel to the strike (130° —140°), indicating a sedimentary fabric. Therefore, we believe that the compaction is probably one of the factors that induce lower inclination of terrestrial sediments of north Tarim than that of marine sediments of SW Tarim.

Numerical experiments show that faster sedimentation rates can produce more inclination shallowing than slower rates. The Lower Cretaceous strata in north Tarim is incompletely developed with a lack of the top part of the Lower Cretaceous, the sedimentary rocks reach a thickness of 1600 m[21]. Comparatively, the Cretaceous marine strata of SW Tarim are well developed with a total thickness of 1600 m, and the Lower Cretaceous stratum is 1000.85 m thick[1]. We can easily deduce that sedimentary rate of the Cretaceous strata of north Tarim is much faster than that of SW Tarim, which again may lead to a much shallower inclination in north Tarim.

However, the inclination shallowing also occurs in marine strata. One example is that the Cretaceous marine Ladd Formation and Point Loma Formation in south California terrain show about 20° inclination shallowing [21]. No indication of magnetic anisotropy may also show shallowing [7]. In comparison with volcanic rocks and Eurasian APWP, the Cretaceous marine sediments of this study may possibly show lower inclination.

Acknowledgements This project was supported in part by the National Natural Science Foundation of China (Grant Nos. 40274014), and by Tarim Oil Field Company, PetroChina Company Limited (Contract No. 41301091008).

References

1. Hao, Y. C., Guo, X. P., Ye, L. S. et al., The Boundary Between the Marine Cretaceous and Tertiary in the Southwest Tarim Basin (in Chinese), Beijing: Geological Publishing House, 2001, 1—23.
2. Yang, Z. Y., Yin, J. Y., Sun, Z. M. et al., Discrepant Cretaceous paleomagnetic poles between Eastern China and Indochina: a consquence of the extrusion of Indochina, Tectonophysics, 2001, 334: 101—113.
3. Yang, Z. Y., New Mesozoic apparent polar wander path for south China: tectonic consequences, J. Geophys. Res., 2001, 106(5): 8493—8520.
4. Meng, Z. F., Deng, Y. S., Ding, Z. H. et al., New paleomagnetic results from Ceno-Mesozoic volcanic rocks along southern rim of the Tarim Basin, China, Science in China, Ser. D, 1998, 41(supp.): 91—104.
5. Gilder, S., Chen, Y., Cogné, J. P. et al., Paleomagnetism of Upper Jurassic to Lower Cretaceous volcanic and sedimentary rocks from the western Tarim Basin and implications for inclination shallowing and absolute dating of the M-0 (ISEA?) chron, Earth Planet Sci. Lett., 2003, 206: 587—600.
6. Zhu, R. X., Yang, Z. Y., Wu H. N. et al., Paleomagnetic constraints on the tectonic history of the major blocks of China during the Phanerozoic, Science in China, Ser. D, 1998, 41(supp.): 1—15.
7. Huang, B. C., Wang, Y. C., Zhu, R. X., New paleomagnetic results and magnetic fabrics of Early Cretaceous rocks in the intermontane basin of Turban, Science in China, Ser. D, 2003, 33(4): 362—372.
8. Lowrie, W., Identification of ferromagnetic minerals in a rock coercivity and unblocking temperature properties, Geophys. Res. Lett, 1990, 17: 159—162.
9. Cognè, J. P., PaleoMac: a MacintoshTM application for treating paleomagnetic data and making plate reconstruction, Geochem Geophys Geosyst, 2003, 4(1), 1007, doi:10.1029/2001GC000227.
10. Zijderveld, J. D. A., AC demagnetization of Rocks: Analysis of Results (eds. Collinson, D. W., Creer, K. M., Runcorn, S. K.), Methods on Paleomagnetic, New York: Elsevier, 1967, 245—286.
11. Kirschvink, J. K., The least-squares line and plane and the analysis of paleomagnetic data, Geophys. J. R. Astron. Soc., 1980, 62: 699—718.
12. McFadden, P. L., McElhinny, M. W., Classification of the reversal test in paleomagnetism, Geophys. J .Int., 1990, 103: 725—729.
13. Fang, D. J., Shen, Z. Y., Tan, X. D., Early Cretaceous paleomagnetic result from Kuche Depress, Tarim Block and its inclination shallowing, Chinese Journal of Geophysics (in Chinese), 2001,

44(1): 72—82.
14. Fang, D. J., Wang, P. Y., Shen, Z. Y., Cenozoic paleomagnetic results and Phanerozoic apparent polar wandering path of Tarim Block, Science in China, Ser. D, 1998, 41(supp.): 105—112.
15. Deng, Q. D., Zhang, P. Z., Rang, Y. K. et al., Basic characteristics of active tectonic of China, Science in China, Ser. D, 2003, 46(4): 356—372.
16. Besse, J., Courtillot, V., Apparent and true polar wander and geometry of the geomagnetic field over the last 200 Myr, J. Geophys. Res., 2002, 107(B11), 2300, doi: 10.1029/2000JB000050.
17. Fang, D. J., Shen, Z. Y., Tan, X. D., Rock Anisotropy of Isothermal Remanent Magnetization and Its Application in Correction of Inclination Shallowing, Chinese Journal of Geophysics (in Chinese), 2000, 43(5): 719—724.
18. Bazhenov, M. L., Mikolaichuk, A. V., Paleomagnetism of Paleogene basalts from the Tien Shan, Kyrgyzstan: rigid Eurasia and dipole geomagnetic field, Earth Planet Sci. Lett., 2002, 195: 155—166.
19. Tan, X. D., Kodama, K. P., Chen, H. L. et al., Paleomagnetism and magnetic anisotropy of Cretaceous red beds from the Tarim basin, northwest China: Evidence for a rock magnetic cause of anomalously shallow paleomagnetic inclinations from central Asia, J Geophys Res, 2003, 108(B2), 2107, doi: 10.1029/2001JB001608.
20. Zhou, Z. Y., Stratigraphy of the Tarim Basin (in Chinese), Beijing: Science Press, 2001, 261—324.
21. Tan, X. D., Kodama, K. P., Compaction-corrected inclinations from southern California Cretaceous marine sedimentary rocks indicate no paleolatitudinal offset for the Peninsular Ranges terrane, J. Geophys. Res., 1998, 103: 27169—27192.